The no-four-concyclic variant V4: a compact one-lemma proof

Companion note

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Abstract

For $n \geq 4$, consider n distinct real planar points which are not all collinear and no four of which lie on one proper Euclidean circle. Count the distinct proper circles determined by selected noncollinear triples. We prove that the minimum is

$$\binom{n-1}{2},$$

and that equality occurs precisely for a near-pencil: $n-1$ points on a line and one point off it. The proof identifies circles with noncollinear triples and applies one elementary richest-line count. It uses no finite search, classification, induction, or external incidence theorem.

1 The frozen variant and the result

Definition 1.1 (V4 configuration)**.** A *V4 configuration* is a finite set $P \subset \mathbb{R}^2$ of $n \geq 4$ distinct points such that P is not collinear and no proper Euclidean circle contains four points of P. A circle is counted if it contains a noncollinear triple from P; it is counted once, independently of its full selected trace. Write $C(P)$ for this count and $f_{\mathrm{V4}}(n)$ for its minimum.

The condition *not collinear* is part of the definition. Without it, a collinear set would determine no proper circle and the minimum would be zero. Lines themselves are never counted.

Definition 1.2 (Near-pencil)**.** A noncollinear n-point set is a *near-pencil* if some line contains exactly $n-1$ of its points.

Theorem 1.3. *For every integer $n \geq 4$,*

$$\boxed{f_{\mathrm{V4}}(n) = \binom{n-1}{2}}. \tag{1}$$

Moreover, a V4 configuration attains this value if and only if it is a near-pencil. In particular, the values for $n = 4, \ldots, 10$ are

$$3, 6, 10, 15, 21, 28, 36.$$

2 The elementary circle interface

We record the entire geometric input, so that the later extremal argument is purely combinatorial.

Lemma 2.1. *Three distinct noncollinear real points determine a unique proper Euclidean circle. A proper circle and a line have at most two common points.*

Proof. A circle has an equation

$$x^2 + y^2 + Ax + By + D = 0. \tag{2}$$

Substitution of three points gives a linear system for A, B, D whose coefficient determinant is twice their signed triangle area. It is nonzero exactly when the points are noncollinear. Thus the solution is unique; because the equation contains three distinct real points, its radius is positive.

After a rigid change of coordinates, a given line is $y = 0$. Restricting (2) to it gives $x^2 + Ax + D = 0$, which has at most two real roots. This proves the intersection bound. ☐

Let $\tau(P)$ denote the number of noncollinear three-subsets of P.

Corollary 2.2 (Triple–circle bijection)**.** *For every V4 configuration,*

$$C(P) = \tau(P). \tag{3}$$

Proof. By lemma 2.1, every noncollinear triple determines one proper circle. If two distinct triples determined the same circle, their union would contain at least four distinct selected points on that circle, contrary to definition 1.1. The map is therefore injective. It is surjective because a counted circle is, by definition, one containing a selected noncollinear triple. ☐

3 One extremal lemma

The next statement no longer mentions circles and is valid for every finite noncollinear real point set.

Lemma 3.1 (Richest-line triple bound)**.** *Let $P \subset \mathbb{R}^2$ be a set of $n \geq 4$ distinct points which are not all collinear. Then*

$$\tau(P) \geq \binom{n-1}{2}. \tag{4}$$

Equality holds if and only if P is a near-pencil.

Proof. Choose a line L containing the largest possible number m of points of P, and put

$$k = n - m = |P \setminus L|.$$

A line through a selected pair shows $m \geq 2$, while noncollinearity gives $k \geq 1$.

First choose two points of $P \cap L$ and one point outside L. These are exactly

$$k\binom{m}{2}$$

noncollinear triples. Next fix an unordered pair $x, y \in P \setminus L$. The line $xy \neq L$ contains at most one selected point of L. Hence at least $m - 1$ points of $P \cap L$ form a noncollinear triple with x, y. Different outside pairs give different triples, so this supplies at least

$$\binom{k}{2}(m-1)$$

further triples. The two families are disjoint, since their members contain respectively two and one points of L. Triples wholly outside L may be omitted. Consequently

$$\begin{aligned} \tau(P) &\geq k\binom{m}{2} + \binom{k}{2}(m-1) \\ &= \frac{k(m-1)(n-1)}{2}. \end{aligned} \tag{5}$$

The exact excess of the displayed lower bound over the target is

$$\frac{k(m-1)(n-1)}{2} - \binom{n-1}{2} = \frac{(n-1)(k-1)(m-2)}{2} \geq 0. \tag{6}$$

This proves (4).

Suppose equality holds. The sandwich formed by (5)–(6) forces $k = 1$ or $m = 2$. In the first case L contains $n - 1$ points, so P is a near-pencil. In the second case maximality of L says that no three points of P are collinear. Every triple is then noncollinear, and

$$\tau(P) - \binom{n-1}{2} = \binom{n}{3} - \binom{n-1}{2} = \frac{(n-1)(n-2)(n-3)}{6} > 0,$$

contrary to equality for $n \geq 4$. Conversely, in a near-pencil the noncollinear triples are exactly the outsider together with a pair from the long line, so there are $\binom{n-1}{2}$. This completes the equality classification. □

Remark 3.2. The richest-line choice is needed only for the equality classification; the lower count itself works for any line determined by two selected points. For $n \geq 4$, the $(n-1)$-point line of a near-pencil is unique: two such distinct lines would contain at least $2n - 3 > n$ selected points in their union. Thus the equality statement classifies the incidence type, not a unique metric realization.

4 Proof and construction

For a V4 configuration, corollary 2.2 and lemma 3.1 give immediately

$$C(P) = \tau(P) \geq \binom{n-1}{2}.$$

Construction 4.1 (Near-pencil extremizers)**.** Take

$$(0,0), (1,0), \ldots, (n-2,0), \qquad u = (0,1).$$

The set is not collinear. By lemma 2.1, a proper circle contains at most two selected points of the line $y = 0$; there is only one selected point outside that line. Thus every proper circle contains at most three selected points, and the construction is V4-admissible.

Its noncollinear triples are precisely u together with a pair on $y = 0$. Therefore corollary 2.2 gives exactly $\binom{n-1}{2}$ counted circles.

The construction proves the upper bound in (1), while the preceding inequality proves the lower bound. If a V4 configuration attains equality, lemma 3.1 makes it a near-pencil; conversely, the same construction argument applies to every near-pencil. This proves theorem 1.3.

5 Scope and computational boundary

The theorem concerns the frozen variant in definition 1.1. It is not the historical unrestricted Erdős problem: the no-four-concyclic condition is an additional hypothesis. It is also distinct from the no-three- collinear variant, since V4 permits arbitrarily rich lines.

The same formula holds at $n = 3$ if noncollinearity is required: one triangle determines one circle. If instead no three points are allowed to be collinear, then every triple in a V4 configuration determines a different circle, so the count is $\binom{n}{3}$. Such configurations exist for every finite n: add points successively outside the finitely many lines and circles forbidden by the points already chosen.

No computation is a premise of this proof. The companion programs `verify_v4_minimal_lemma.py` and `verify_v4_minimal_lemma_redteam.py` check the polynomial identities (5)–(6) coefficientwise, without looping over n, line sizes, point sets, triples, circles, or configurations. They do not formalize the elementary geometric facts in lemma 2.1. Earlier finite-window and coordinate checks are redundant archival tests, not proof dependencies. This note is not a Lean or Coq formalization.